Class Gastropoda

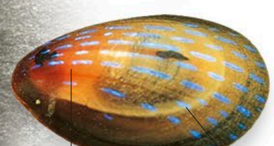
Gastropods

Occurrence c.75,000 spp. worldwide; in aquatic habitats (mostly bottom-dwelling), in wet, damp, and dry regions on land

Habitat All

This vast and variable group of invertebrates includes over four-fifths of the mollusks alive today. The familiar garden snail, slug, winkle, sea hare, sea slug, abalone, pond snail, and conch are all gastropods. Typical species have a spiral shell and move by creeping on a muscular foot, but there are many exceptions to this rule: some,for example, have no shell at all. Other characteristic features include a large head, with eyes and tentacles, and a ribbon-shaped mouthpart called a radula, which is armed with small, often microscopic, teeth and is used to rasp away at food. These mollusks vary widely in size, and can grow up to 3¼ ft (1 m) long. The shape of the shell is highly variable—some form a tall spire, while others flare outward. In some species, an operculum, or "door," shuts the shell opening when the animal withdraws inside. Gastropods are characterized by torsion, a process in which the mantle cavity is rotated counterclockwise up to 180 degrees, until it faces forward and is positioned over the head. This occurs early in larval development and means that the digestive, reproductive, and excretory systems all discharge through the one opening in the shell. Gastropods feed in a wide variety of ways: they may be hunters, browsers, or grazers; some are suspension or deposit feeders; some are parasitic. Although sexes are separate in most gastropods, several groups are hermaphrodites, while in other groups, animals change from one sex to the other during their lifetime.

SEA HARES
Most sea hares have a thin shell (not visible externally), a large, well-developed head with 2 pairs of tentacles, and a large foot. Sea hares inhabit beds of seaweed and sea grasses, where they feed on large, fleshy algae. The North Atlantic species shown above, Aplysia punctata, *may attain a length of more than 12 in (30 cm).*



LIMPETS
Found on rocky shores, limpets, like abalones, graze on young algae, using their conical shells to protect themselves from the waves. Of the 2 Atlantic species shown here, the common limpet, Patella vulgata *(left)— 2 in (5 cm) across— wanders over rocks, while the blue-rayed limpet,* Ansates pellucida *(below)—up to ¾ in (2 cm) across—creeps over kelp, feeding on small epiphytic algae and on the kelp itself. Many rock-dwelling limpets are strongly territorial, making circular scars in the rock where they stay locked in position at low tide.*

HELMET SHELLS
The stout, colorful, and attractive helmet shells resemble the helmets worn by the gladiators of ancient Rome. The shells may be more than 12 in (30 cm) long. The animals that make them inhabit sandy sediments in the tropics, feeding on other mollusks and sea urchins. The shells of species such as the horned helmet, Cassis cornuta *(left), are much used as ornaments and for carving.*

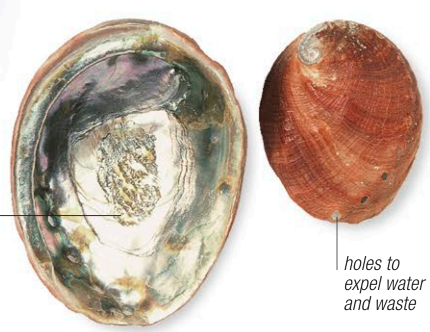
MUREXES
These gastropods are predators, often consuming other mollusks and barnacles after boring a hole through their shells. Several secrete a purple substance that has been used for centuries as a dye. Murex troscheli *(right), which is found in the Indo-Pacific, has a shell that is 6 in (15 cm) long, with elongated, spiny extensions.*

VIOLET SEA SNAIL
The violet sea snail, Janthina janthina, *is one of the few shelled gastropods that inhabit the ocean surface. It does this by secreting a raft of mucus bubbles that keeps it afloat. This warm-water snail is blind and has a paper-thin shell, ¾ in (2 cm) across. It feeds on by-the-wind-sailors (see p.540) and other colonial hydroids that drift through the water.*

COWRIES
The beauty of their glossy shells ensures that the mostly tropical cowries are particularly sought after by collectors. In life, the shell is covered by a flap of skin, which is a part of its brightly colored mantle. The shell aperture is long and narrow, which helps to protect the cowrie from attacks by predatory crabs. This species, Cypraea ocellata, *can grow up to 2¼ in (6 cm) across.*

HARP SHELLS
The globular harp shells have smooth, regularly spaced ribs, which resemble the strings of a harp. Up to 4 in (10 cm) long, the shells are often brown, red, or pinkish. Species such as Harpa major, *shown here, are mainly nocturnal, and inhabit tropical sandy sediments, into which they burrow. When disturbed, they may discard part of the large foot, in much the same way as a lizard discards its tail.*

CONE SHELLS
Cone shells are highly dangerous predators: using their harpoonlike mouthparts, they attack fish and other animals, injecting a venom so potent that it has been known to cause death in humans. The West African garter cone, Conus genuanus, *shown here, is a typical example, with a highly decorated, tapering shell up to 2 ¾ in (7 cm) long.*



ABALONES
Also known as ormers, awabis, or ear-shells, abalones are grazing animals that can be over 8 in (20 cm) across. Their shells have a distinctive series of holes, through which the water current drawn into the gill chamber leaves, as do waste products. The iridescent shells of species such as the red abalone, Haliotis rufescens *(above), found off the west coast of North America, are used to make jewelry, and the animals inside are often eaten.*